

Silhouette Score Explained with an Example

The **Silhouette Score** measures how well data points fit into their assigned clusters.

It answers two questions:

1. **How close is a point to other points in its own cluster?** (Cohesion)
2. **How far is it from points in the nearest neighboring cluster?** (Separation)

This makes it one of the best methods for determining the **optimal number of clusters (K)**.

Formula

For each point:

a = Average distance to points in the same cluster
 b = Average distance to points in the nearest cluster

Silhouette Score:

$$s = \frac{b - a}{\max(a, b)}$$

Where:

- **a** = intra-cluster distance (how tightly grouped)
 - **b** = nearest-cluster distance (how far from other clusters)
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Example

Suppose we have two clusters:

Cluster A

A1 = 1
A2 = 2
A3 = 3

Cluster B

B1 = 10
B2 = 11
B3 = 12

Visual:

1 2 3 10 11 12
|---Cluster A---| |---Cluster B---|

Let's calculate the Silhouette Score for point **A2 = 2**.

Step 1: Calculate a (Within-Cluster Distance)

Distance from A2 to other points in Cluster A:

Distance(A2,A1) = |2-1| = 1
Distance(A2,A3) = |2-3| = 1

Average:

$$a = \frac{1 + 1}{2}$$
$$a = 1$$

Step 2: Calculate b (Nearest Cluster Distance)

Distances from A2 to Cluster B:

Distance(2,10)=8
Distance(2,11)=9
Distance(2,12)=10

Average:

$$b = \frac{8 + 9 + 10}{3}$$
$$b = 9$$

Step 3: Calculate Silhouette Score

$$s = \frac{9 - 1}{9}$$

$$s = 0.889$$

Result

Silhouette Score = 0.889

This is very close to 1, meaning A2 is correctly clustered.

Example of a Poorly Clustered Point

Suppose:

Cluster A: 1, 2, 8
 Cluster B: 10, 11, 12

Point:

8

looks closer to Cluster B.

Within Cluster

Distance(8, 1) = 7
 Distance(8, 2) = 6

Average:

$$a = 6.5$$

To Cluster B

Distance(8, 10) = 2
 Distance(8, 11) = 3
 Distance(8, 12) = 4

Average:

$$b = 3$$

Silhouette:

$$s = \frac{3 - 6.5}{6.5}$$

$$s = -0.54$$

Result

Negative score!

This indicates the point may belong to the other cluster.

Interpretation

Score	Meaning
+1	Perfect clustering
0.7 to 1.0	Excellent
0.5 to 0.7	Good
0.25 to 0.5	Weak
Around 0	Overlapping clusters
Negative	Wrong cluster assignment

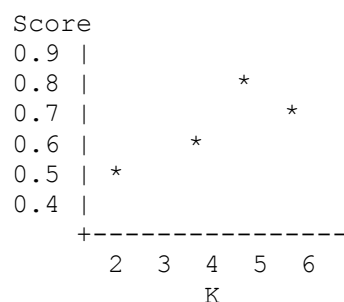
Using Silhouette Score to Find K

Suppose we run K-Means with different K values.

K Silhouette Score

2 0.52
 3 0.67
 4 0.81
 5 0.72
 6 0.63

Graph:



The highest score occurs at:

$$K = 4$$

Therefore, 4 is the optimal number of clusters.

Real-Life Example: Customer Segmentation

Suppose a retailer segments customers by:

- Annual Income
- Spending Score

$$K = 2$$

Low Spenders
High Spenders

$$\text{Silhouette} = 0.58$$

$$K = 3$$

Budget Customers
Regular Customers
Premium Customers

$$\text{Silhouette} = 0.79$$

$$K = 5$$

Too many overlapping groups.

$$\text{Silhouette} = 0.61$$

Since $K = 3$ has the highest score, it is chosen.

Elbow Method vs Silhouette Score

Feature	Elbow Method	Silhouette Score
Uses WCSS	Yes	No
Measures Separation	No	Yes
Measures Cohesion	No	Yes

Feature	Elbow Method	Silhouette Score
Objective	Partly subjective	More objective
Accuracy	Good	Better

Interview Answer

Silhouette Score evaluates the quality of clustering by measuring how similar a data point is to its own cluster compared to other clusters. It ranges from -1 to +1. A value close to +1 indicates well-separated clusters, a value near 0 indicates overlapping clusters, and a negative value suggests incorrect cluster assignment. To determine the optimal number of clusters, we compute the average silhouette score for different K values and choose the K with the highest score.

Quick Memory Trick

a = Same Cluster Distance
b = Nearest Other Cluster Distance

If $b \gg a$ → Good Clustering
If $b \approx a$ → Overlap
If $b < a$ → Wrong Cluster

That's why the Silhouette Score is often preferred over the Elbow Method when selecting the optimal number of clusters.